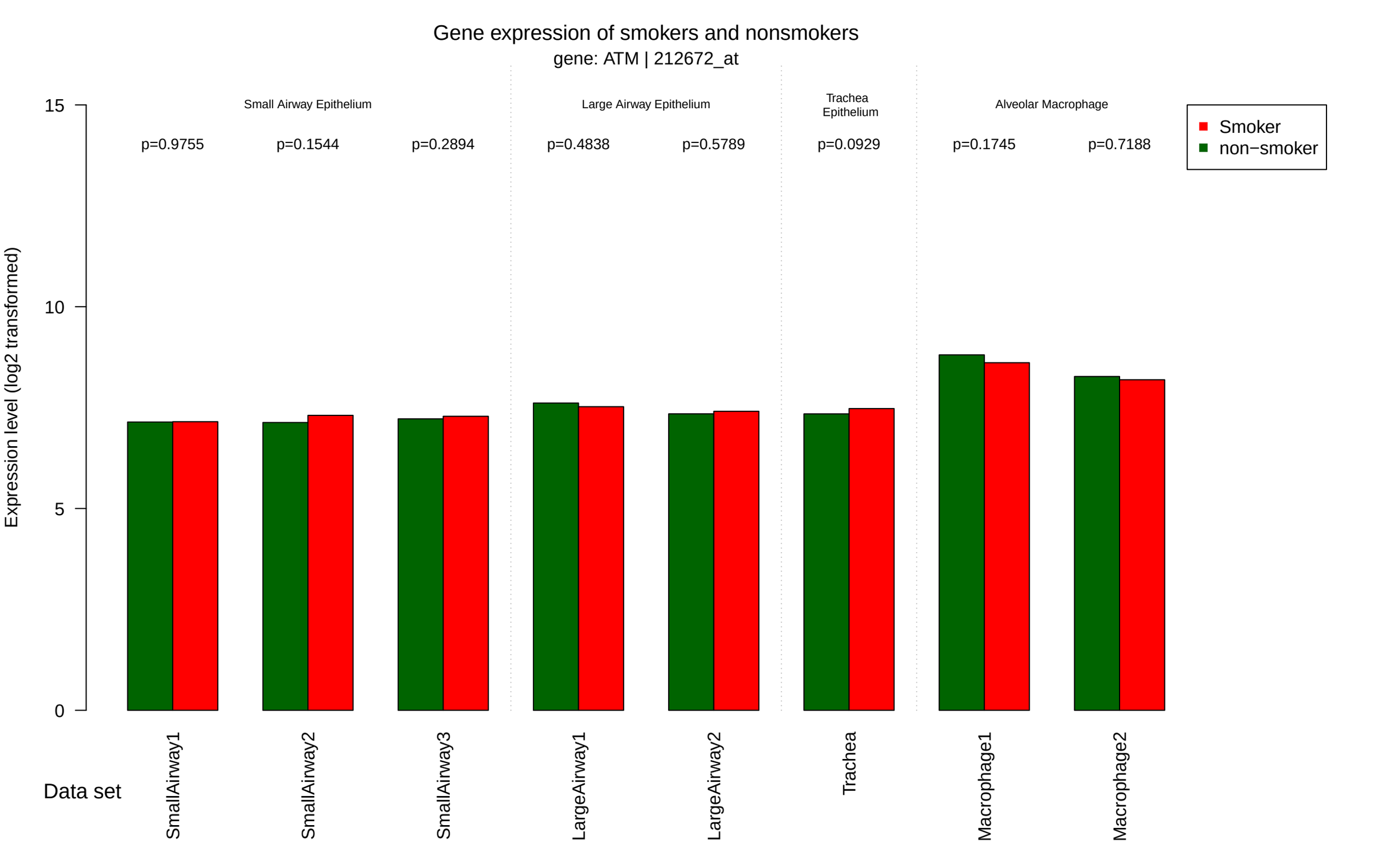
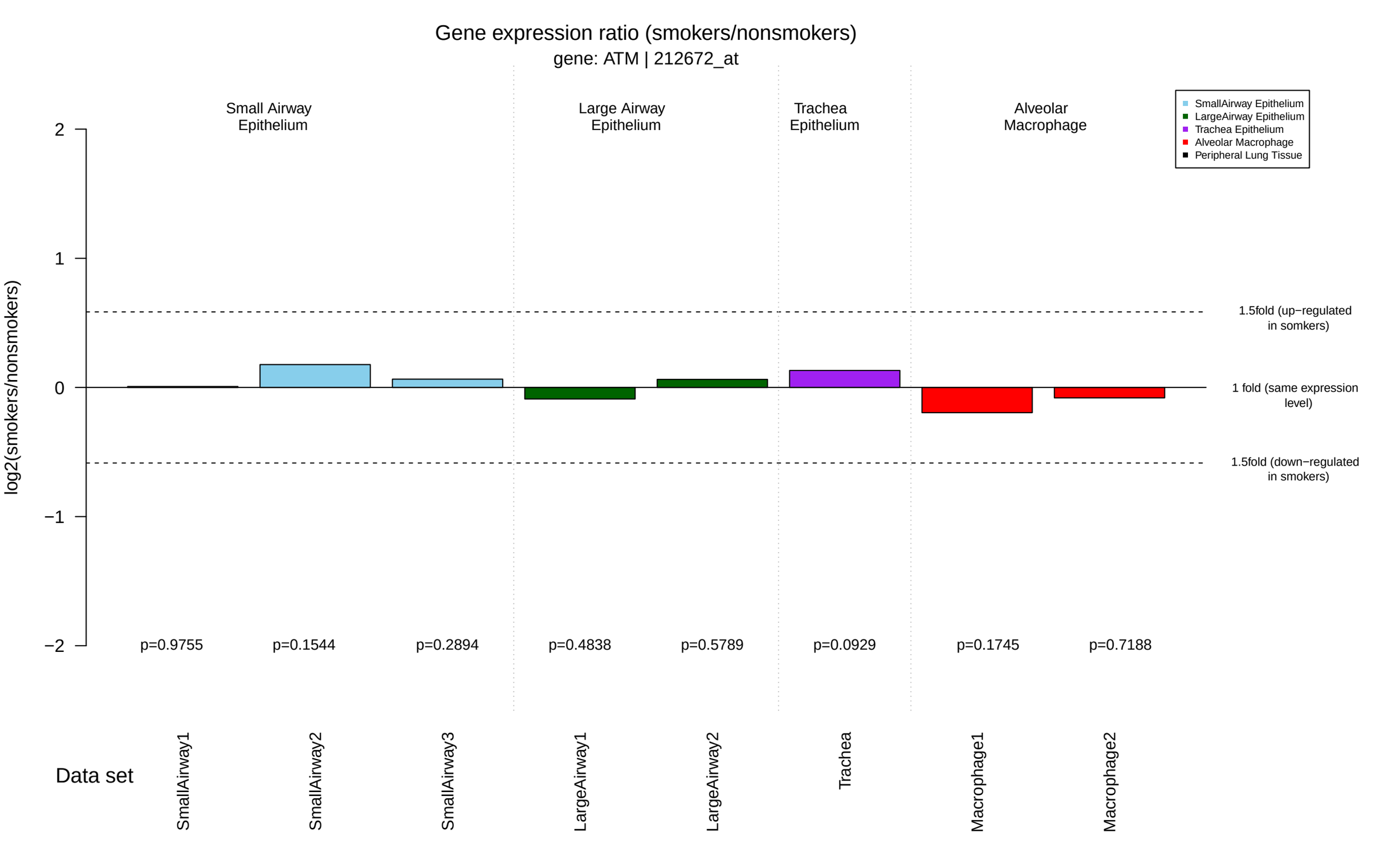
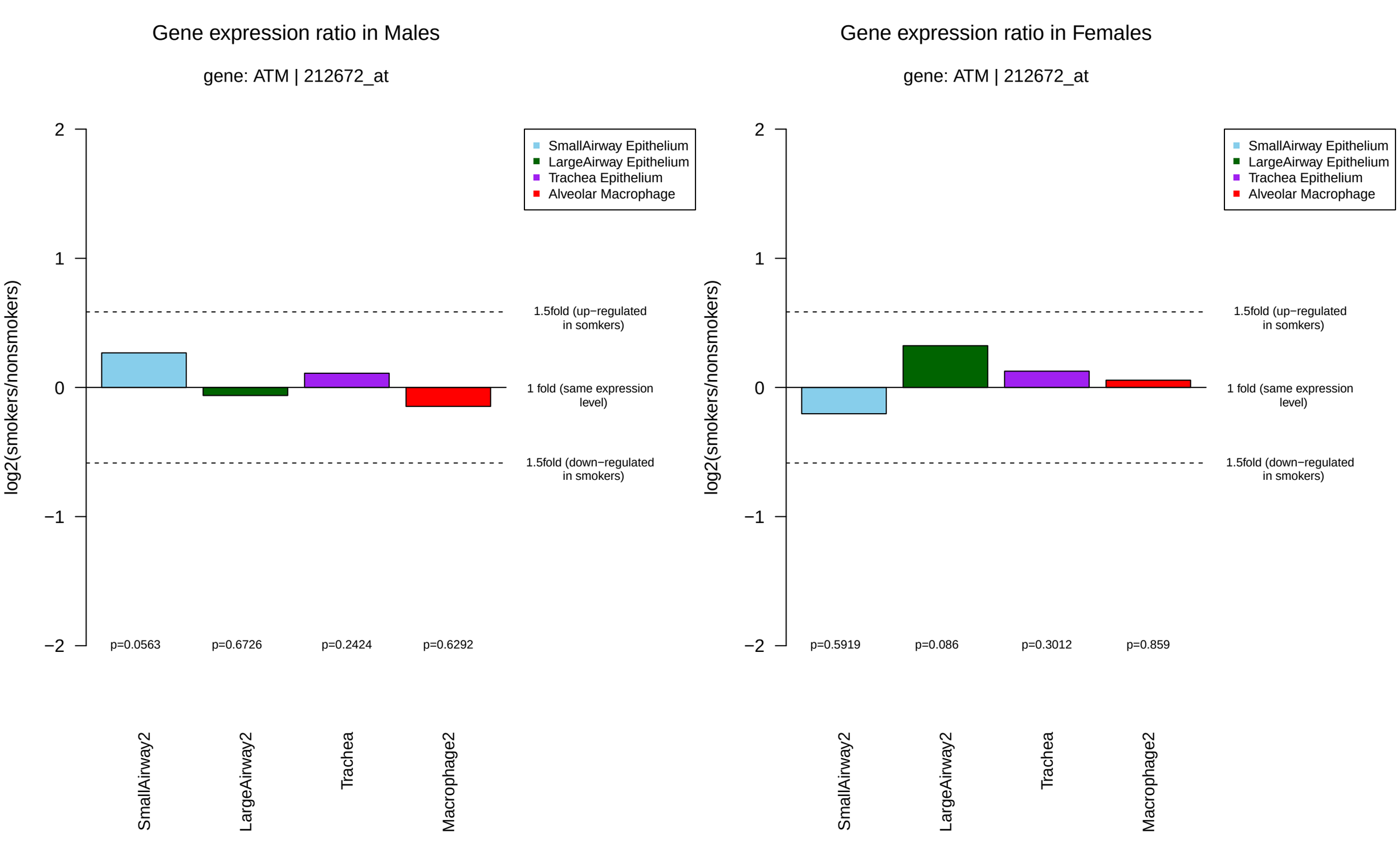
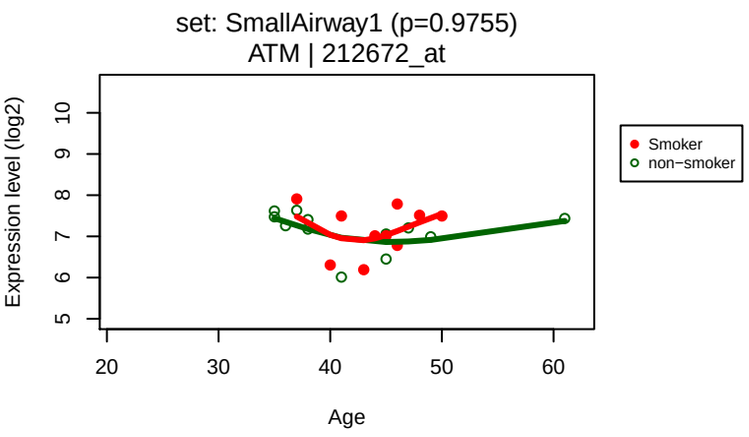
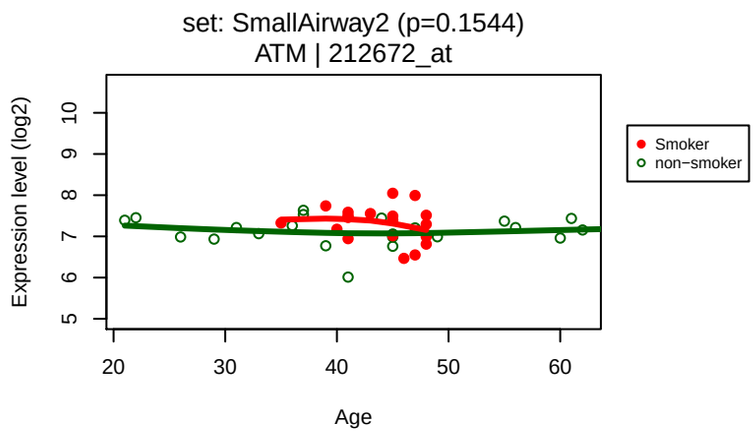
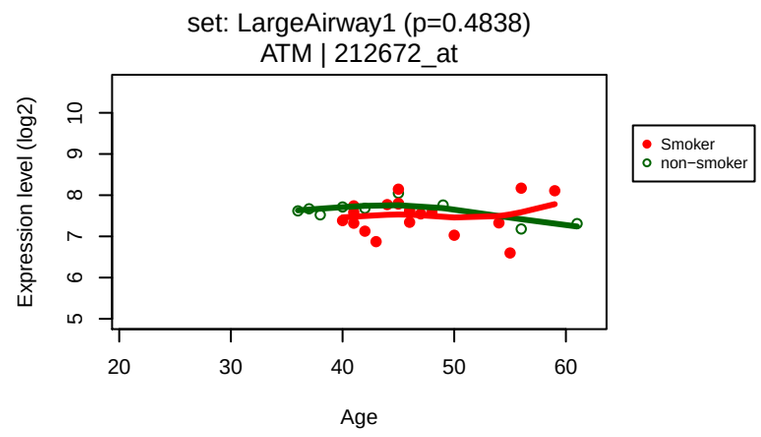

Gene expression of smokers and nonsmokers
gene: ATM | 212672_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9755
p=0.1544
p=0.2894
p=0.4838
p=0.5789
p=0.0929
p=0.1745
p=0.7188
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

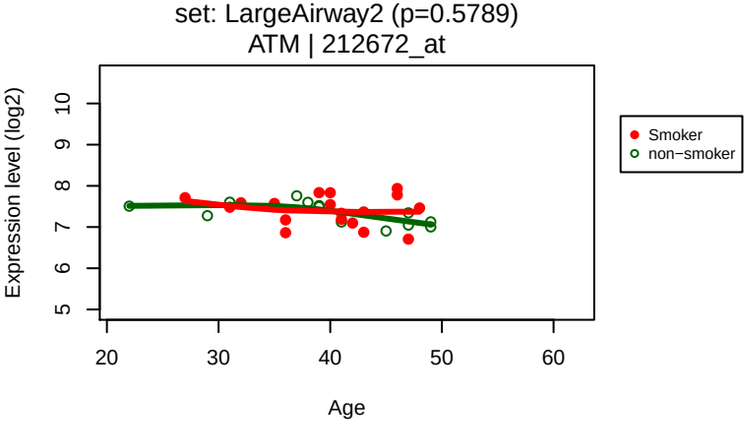

Gene expression ratio (smokers/nonsmokers)
gene: ATM | 212672_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9755
p=0.1544
p=0.2894
p=0.4838
p=0.5789
p=0.0929
p=0.1745
p=0.7188
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

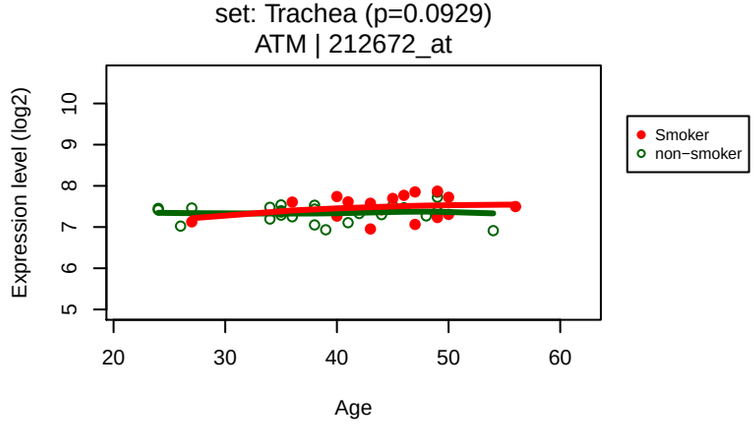

Gene expression ratio in Males
gene: ATM | 212672_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0563
p=0.6726
p=0.2424
p=0.6292
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATM | 212672_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5919
p=0.086
p=0.3012
p=0.859
SmallAirway2
LargeAirway2
Trachea
Macrophage2

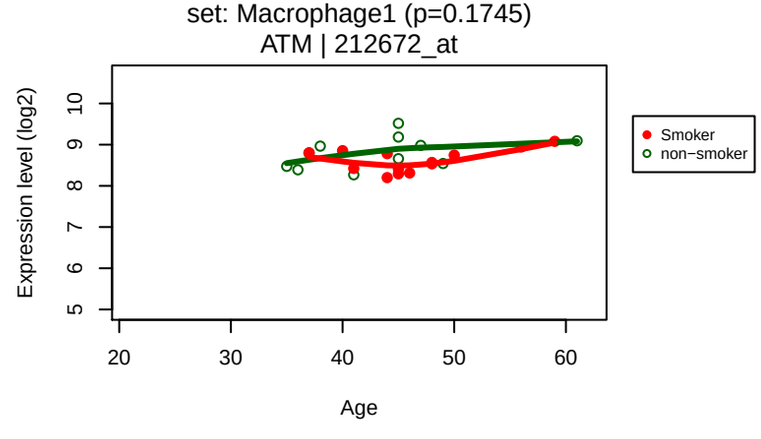

set: SmallAirway1 (p=0.9755)
ATM | 212672_at
Expression level (log2)
Age
Smoker
non−smoker

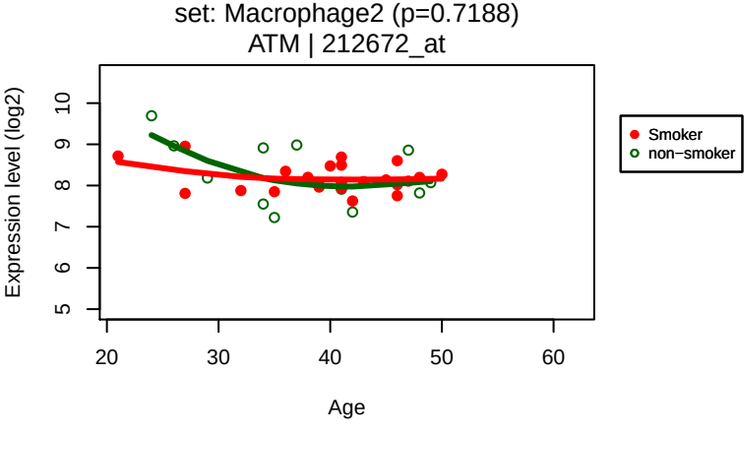

set: SmallAirway2 (p=0.1544)
ATM | 212672_at
Expression level (log2)
Age
Smoker
non−smoker

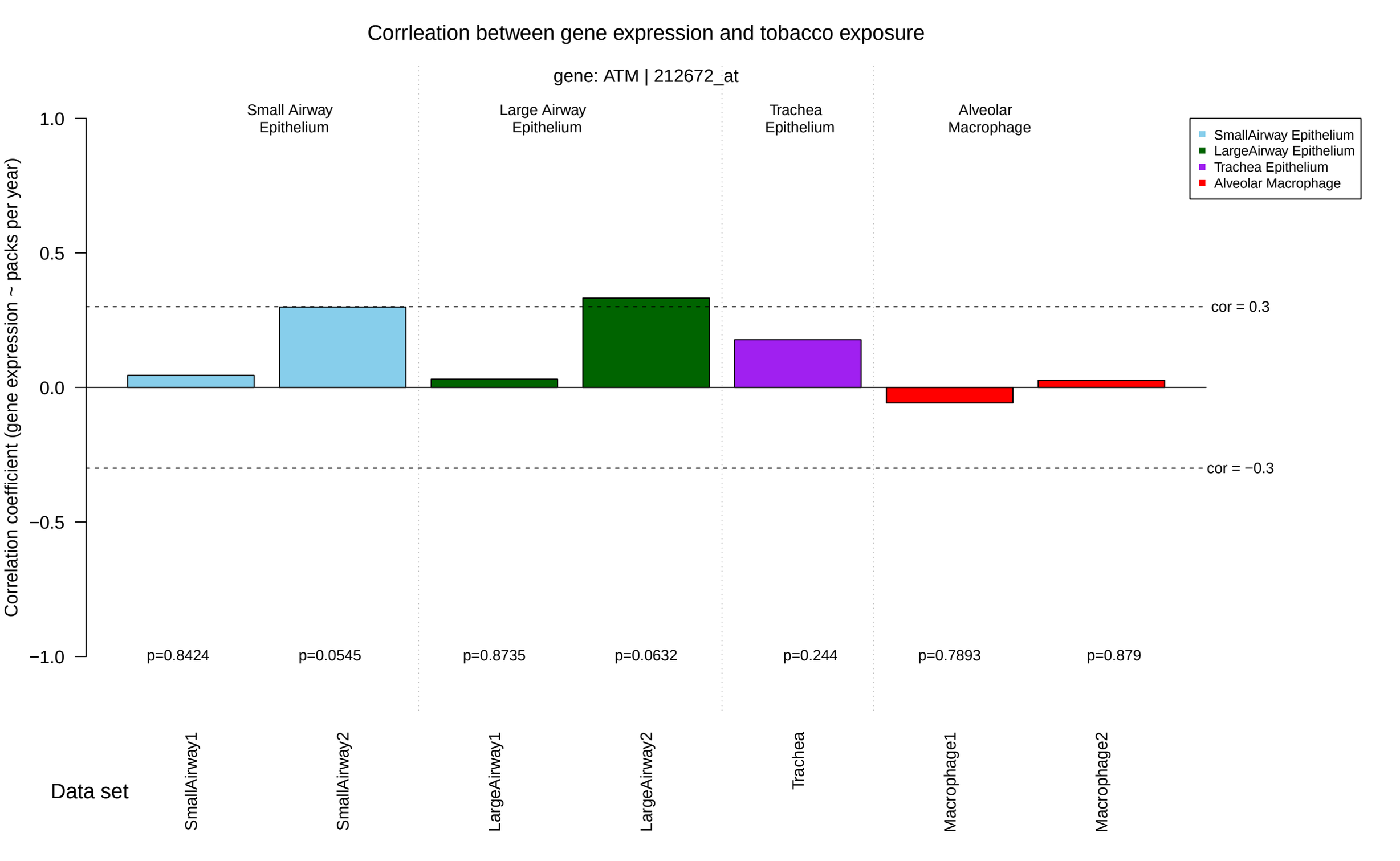

set: LargeAirway1 (p=0.4838)
ATM | 212672_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5789)
ATM | 212672_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0929)
ATM | 212672_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1745)
ATM | 212672_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7188)
ATM | 212672_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATM | 212672_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8424
p=0.0545
p=0.8735
p=0.0632
p=0.244
p=0.7893
p=0.879
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2